



When I bought my first IoT-enabled light bulb and controlled it with my phone about five years ago, I thought it was the coolest thing and that I would cherish it for a long time. That thought changed pretty soon, though. Today I have over 300 IoT devices at home, which are controlled with numerous apps. I manage these by using a spreadsheet that maps each device to its app so I don't go crazy having to memorise all the device controls. I sincerely hope advancements in IoT architecture will enable simpler and integrated ways of managing devices.

IoT architecture comprises the numerous elements that make up an IoT networking system like sensors, protocols, actuators, cloud services, edge services, data services, and applications. The basic three-layered architecture is easy to implement. It helps to collect data, process it, and then transform it into usable

information. Embedded analytics with machine learning and artificial intelligence enables businesses to remain agile. The three layers are:

- *Perception* for the sensors, gadgets, and other devices
- *Network* for the connectivity between devices and applications
- *Application* for the user interaction and responses

IoT architecture makes all this possible by ensuring that data is handled properly and gets to the right place. Networks would become unstable without adequate IoT architecture, undermining the very point of investing in IoT in the first place.

Although this 3-layer architecture identifies key layers and is the simplest to implement, it does leave out details that are critical in large-scale enterprise-grade IoT solutions, especially security threat concerns needing specific attention across each of the layers. This has led to the

evolution of the IoT architecture into 4, 5, 6, and 7-layer architectures.

Four-layer architecture is like the basic 3-layer architecture with one key additional layer breaking down the network connectivity layer and creating a new support layer. Adding the new support layer between the connectivity layer and the perception layer improves the ability to manage data efficiently and its security.

The 5-layer architecture is one of the most adopted and implemented architectures, because of its simplicity. It gives the right amount of balance, allowing greater management complexity along with the ability to scale the number of IoT devices in use. Here, the collaboration layer is separated from the application layer. This makes IoT devices much more lightweight, as the processing of data is moved into the application layer that could be deployed on the cloud or edge computing.